An Optimal Estimate for the Local Discontinuous Galerkin Method

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Abstract. L^2 error estimates for the Local Discontinuous Galerkin (LDG) method have been theoretically proven for linear convection diffusion problems and periodic boundary conditions. It has been proven that when polynomials of degree k are used, the LDG method has a suboptimal order of convergence k. However, numerical experiments show that under a suitable choice of the numerical flux, higher order of convergence can be achieved. In this paper, we consider Dirichlet boundary conditions and we show that the LDG method has an optimal order of convergence $k+1$.

1 Introduction

Over the last decade, the Runge Kutta Discontinuous Galerkin (RKDG) method introduced and analyzed by Cockburn and Shu, see [CS98,CSK], has become a practical method for solving hyperbolic systems. The RKDG method can provide high order of approximation and has a high degree of parallelism. Originally, it was designed for solving nonlinear hyperbolic systems. Recently, Cockburn and Shu, [CS98], have extended the RKDG method to time-dependent convection diffusion systems. The extension proposed in [CS98], the Local Discontinuous Galerkin (LDG) method, can also provide high order accuracy and, because of its local nature, is highly parallelizable. In [CS98], it was shown that, for linear convection diffusion problems and periodic boundary conditions, the method is at least of order k when polynomials of degree k are used. In this paper, we show that, under a suitable choice of the numerical flux, an optimal order $k+1$ can be achieved even for Dirichlet boundary conditions.

We now describe the general formulation of the LDG method, applied to the following Cauchy problem

$$\partial_t u + \partial_x \left(f(u) - a(u)\partial_x u \right) = 0 \quad u(0,x) = u_0. \tag{1}$$

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First, the equation is rewritten in conservation form. We obtain the following system

$$\begin{aligned} \partial_t u + \partial_x \left(f(u) - \sqrt{a(u)} q \right) &= 0, \quad \text{in } (0,T) \times \Omega, \\ q - \partial_x g(u) &= 0, \quad \text{in } (0,T) \times \Omega, \\ u(0,x) &= u_0. \quad \text{on } \Omega. \end{aligned} \tag{2}$$

where $q = \sqrt{a(u)}\partial_x u$ and $g(u) = \int^u \sqrt{a(s)} ds$. Then, the system is discretized by adapting the ideas developed in the RKDG method. Let $\{x_{j+\frac{1}{2}}\}_{j=0}^N$ be a partition of the domain Ω, and $I_j = (x_{j-1/2},\ x_{j+1/2})$. At each time $t \in [0,\ T]$, the solution $w = (u, q)$ of the system (2) is approximated by a function $w_h = (u_h, q_h)$ of the finite element space $V_h = V_h^k \times V_h^k$ of piecewise discontinuous functions

$$V_h^k = \{v \in L^1(\Omega) | v_{|I_j} \in P^k(I_j)\}.$$

The restriction of w_h to an interval I_j satisfies, for any test function (v_u, v_q) in V_h, the following weak relations

$$\begin{aligned} \int_{I_j} \partial_t u_h v_u &= \int_{I_j} h_u(w_h) \partial_x v_u - \hat{h}_u(w_h) v_u \Big|_{x_{j-1/2}}^{x_{j+1/2}}, \\ \int_{I_j} q_h v_q &= \int_{I_j} h_q(w_h) \partial_x v_q - \hat{h}_q(w_h) v_q \Big|_{x_{j-1/2}}^{x_{j+1/2}} \end{aligned}$$

where $h(u,q) = (f(u) - \sqrt{a(u)}q, -g(u))$ and $\hat{h}$ is a numerical flux which incorporates the boundary conditions. In our error analysis, we consider the linear convection diffusion equation

$$\partial_t u + \partial_x (cu - a\partial_x u) = 0,\ u(0,x) = u_0,\ u(t,\alpha) = \delta_\alpha(t),\ u(t,\beta) = \delta_\beta(t) \tag{3}$$

where c is an arbitrary, non zero, constant, $a > 0$ and $\Omega = [\alpha,\ \beta]$. For this problem the flux h is given by

$$h(u,q) = \left(cu - \sqrt{a}q, -\sqrt{a}u\right).$$

We consider the following choice of the numerical flux at the interior grid points :

$$\hat{h}\left(w^-, w^+\right) = \begin{cases} \left(cu^- - \sqrt{a}q^+, -\sqrt{a}u^-\right), & \text{if } c > 0, \\ \left(cu^+ - \sqrt{a}q^-, -\sqrt{a}u^+\right), & \text{if } c < 0, \end{cases} \tag{4}$$

With this particular choice an optimal order of convergence can be obtained for the LDG method as described in the following section.

2 Error analysis for linear convection diffusion

In this section, we present an error estimate for the LDG method applied to the linear model problem (3). Our error estimate is based on the technique presented in [CS98]. However, in this paper, we not only obtain an optimal error estimate, but we also consider general Dirichlet boundary conditions.

Theorem 1. *(Error estimate). Let $e(t,x)$ be the error of the approximation obtained by using the LDG method to solve the linear model problem, the quantity defined by,*

$$|||e|||^2 = \| e_u(T) \|^2_{L^2(\Omega)} + \int_0^T \left\{ \| e_q \|^2_{L^2(\Omega)} + |c||e_u(\alpha^+)|^2 + \frac{|c|}{2}|e_u(\beta^-)|^2 \right\}$$

satisfies

$$|||e||| = O\left((\Delta x)^{k+1}\right),$$

provided $u_o \in H^{k+3}$.

Proof. Without loss of generality, we assume a positive convection coefficient, $c > 0$ in (3). The solution w_h obtained from the LDG method satisfies the weak formulation

$$\mathcal{B}_h(w_h, v) = \mathcal{L}_h(v), \quad \forall v \in V_h, \tag{5}$$

where the bilinear form $\mathcal{B}_h$ and the linear operator $\mathcal{L}_h$ are defined as follows

$$\begin{aligned}
\mathcal{B}_h(w,v) = &\int_0^T \int_\Omega \partial_t w_u v_u + \int_0^T \int_\Omega w_q v_q - \int_0^T \sum_1^N \int_{I_j} h(w)^T \partial_x v \\
&- \int_0^T \sum_1^{N-1} \hat{h}(w)^T [v]_{j+1/2} + \int_0^T \left(c w_u(\beta^-) - \sqrt{a} w_q(\beta^-)\right) v_u(\beta^-) \\
&+ \int_0^T \sqrt{a} w_q(\alpha^+) v_u(\alpha^+),
\end{aligned}$$

$$\mathcal{L}_h(v) = \int_0^T \left\{ c\delta_\alpha v_u(\alpha^+) - \sqrt{a}\delta_\alpha v_q(\alpha^+) + \sqrt{a}\delta_\beta v_q(\beta^-) \right\}.$$

Let w be the exact solution of (3). Since w and w_h satisfy (5), we obtain for the error $e = w - w_h$

$$\mathcal{B}_h(e, v) = 0, \quad \forall v \in V_h.$$

Let $\Pi = (\Pi_u, \Pi_q)$ be an arbitrary projection onto $V_h = V^k \times V^k$ and set $v = \Pi(e)$, $p = (p_u, p_q) = \Pi(w) - w$. ¿From the above error equation we get

$$\mathcal{B}_h(v, v) = \mathcal{B}_h(p, v).$$

An error estimate can now be obtained from the previous equation by choosing an appropriate projection Π and by providing an upper bound for the right-hand side. We proceed as follows: First, observe that since the coefficient c is positive, the terms involving the numerical fluxes can be simplified as follows

$$\hat{h}(p)^T[v]_{j+1/2} = \left(cp_u^-[v_u] - \sqrt{a}p_q^+[v_u] - \sqrt{a}p_u^-[v_q]\right)_{j+1/2}.$$

By simply selecting an appropriate projection Π, we can set the above quantity to zero. For the u component, we consider a polynomial that interpolates the function at the right endpoint. For the q component, we consider the left endpoint. We also have $p_u(\beta^-) = p_q(\alpha^+) = 0$. Hence, we obtain the following simplified expression for $\mathcal{B}_h(p,v)$:

$$\int_0^T \int_\Omega \partial_t p_u v_u + p_q v_q - \int_0^T \sum_1^N \int_{I_j} h(p)^T \partial_x v - \int_0^T \sqrt{a} p_q(\beta^-) v_u(\beta^-)$$

Note that in the pure hyperbolic case, $a = 0$, we recover the expression of $\mathcal{B}_h(p,v)$ obtained in [CS98]. An optimal order can be obtained by considering interpolation polynomials at the Gauss-Radau quadrature points. Following the ideas of LeSaint and Raviart [LR74], we define $\Pi_u(f)$ and $\Pi_q(f)$ as the interpolation polynomial on the Gauss-Radau quadrature points, with the appropriate fixed point. From the integration properties of these polynomials and the Bramble-Hilbert lemma, it can be shown that if $f \in H^{k+2}(I_j)$ and $v \in P^k(I_j)$ we have the following estimates

$$\int_{I_j} |(\Pi(f) - f)\, v| \leq C_1(\Delta x)^{k+1} |f|_{H^{k+1}(I_j)} \parallel v \parallel_{L^2(I_j)},$$

$$\int_{I_j} |(\Pi(f) - f)\, \partial_x v| \leq C_2(\Delta x)^{k+1} |f|_{H^{k+2}(I_j)} \parallel v \parallel_{L^2(I_j)}.$$

Applying again the Bramble-Hilbert lemma, we can derive an L^∞ estimate for the error at the boundary endpoint

$$\left|(\Pi(f) - f)\,(\beta^-)\right| \leq C(\Delta x)^{k+1} \parallel f \parallel_{W_\infty^{k+1}(I_j)}.$$

Using these estimates and the regularity of the initial conditions, we obtain the following estimate

$$|\mathcal{B}_h(p,v)| \leq \gamma_1(a,c) C_1(u_0)(\Delta x)^{k+1} \int_0^T \parallel \Pi_u(e) \parallel_{L^2(\Omega)} \tag{6a}$$

$$+ \gamma_2(a) C_2(u_0)(\Delta x)^{k+1} \int_0^T \parallel \Pi_q(e) \parallel_{L^2(\Omega)} \tag{6b}$$

$$+ \gamma_3(a) C_3(u_0)(\Delta x)^{k+1} \int_0^T |\Pi_u(e)(\beta^-)|, \tag{6c}$$

where the constants $\gamma_1(a,c), \gamma_2(a)$ and $\gamma_3(a)$ are such that, $\gamma_1(0,.)$, $\gamma_1(.,0)$, $\gamma_2(0)$ and $\gamma_3(0)$ do not vanish. Applying Young's inequality to the second, (6b), and third, (6c), term we get

$$
\begin{aligned}
|\mathcal{B}_h(p,v)| \leq \rho_1(a,c) C_1(u_0)(\Delta x)^{k+1} \int_0^T \| \Pi_u(e) \|_{L^2(\Omega)} \\
+ \frac{1}{2} \int_0^T \| \Pi_q(e) \|^2_{L^2(\Omega)} + \rho_2(a) \int_0^T \{C_2(u_0)(\Delta x)^{k+1}\}^2 \\
+ \frac{c}{4} \int_0^T |\Pi_u(e(\beta^-))|^2 + \rho_3(a,c) \int_0^T \{(\Delta x)^{k+1} C_3(u_0)\}^2. \quad (7)
\end{aligned}
$$

Since

$$
\begin{aligned}
\mathcal{B}_h(v,v) = \frac{1}{2} \| \Pi_u(e(T)) \|^2_{L^2(\Omega)} - \frac{1}{2} \| \Pi_u(e(0)) \|^2_{L^2(\Omega)} + \int_0^T \| \Pi_q(e) \|^2_{L^2(\Omega)} \\
+ \frac{c}{2} \int_0^T \{ |\Pi_u(e(\alpha^+))|^2 + |\Pi_u(e(\beta^-))|^2 \} + \frac{c}{2} \int_0^T \sum_{j=1}^{N-1} [\Pi_u(e)]^2 \quad (8)
\end{aligned}
$$

combining (7) and (8), we obtain

$$
\begin{aligned}
|||\Pi(e)|||^2 \leq \| \Pi_u(e(0)) \|^2_{L^2(\Omega)} + C_1(\Delta x)^{2(k+1)} \\
+ C_2(\Delta x)^{k+1} \int_0^T \| \Pi_u(e) \|_{L^2(\Omega)},
\end{aligned}
$$

where C_1 and C_2 depend solely on T and u_0. By application of Gronwall's lemma, we finally get

$$
|||\Pi(e)||| = O\left((\Delta x)^{k+1}\right).
$$

From this bound, the error estimate follows. □

3 Numerical experiments

We present here some numerical results, for the LDG method applied to the linear convection diffusion problem (3). The domain is the one dimensional interval $[-1, 1]$ and the initial conditions are $u_o(x) = \sin(x)$. We compute the solution on the time interval $[0, 0.1]$. A sufficiently small fixed time step has been used, such that the contribution to the error is dominated by the spatial error. We present the computed order of convergence for polynomials of degree 1 to 4. In Table 1, we show the L^2-error of the $u-$component at the end time and the computed order for a convection diffusion equation with a small diffusion coefficient $a = 0.01$. In Table 2, the diffusion coefficient a is increased to 1.0. We have also included the L^2 error of u_x at the end time, asserting our claim that an order of $k+1$ is also obtained for the $q-$component.

Table 1. $c = 1.0,\ a = 0.01$

k	$N = 10$	$N = 20$		$N = 40$	
	error	error	order	error	order
1	1.3593e-03	3.5642e-04	1.9313	8.9377e-05	1.9956
2	3.7061e-05	4.4837e-06	3.0471	5.5999e-07	3.0012
3	2.8634e-07	1.6651e-08	4.1040	1.0444e-09	3.9949
4	4.5401e-09	1.3242e-10	5.0996	4.1731e-12	4.9878

Table 2. $c = 1.0,\ a = 1.0$

k		$N = 10$	$N = 20$		$N = 40$	
		error	error	order	error	order
1	u	1.2161e-03	3.1400e-04	1.9534	7.9697e-05	1.9782
	u_x	1.9568e-03	5.0633e-04	1.9503	1.2871e-04	1.9759
2	u	3.9617e-05	4.6240e-06	3.0989	5.4412e-07	3.0872
	u_x	1.9065e-05	2.4675e-06	2.9498	3.1107e-07	2.9878
3	u	2.3411e-07	1.4748e-08	3.9886	9.3217e-10	3.9838
	u_x	3.7879e-07	2.3641e-08	4.0020	1.5160e-09	3.9630
4	u	4.8128e-09	1.3674e-10	5.1374	4.2538e-12	5.0065
	u_x	2.3370e-09	7.7297e-11	4.9181	5.1103e-12	3.9189

4 Conclusion

In this paper we have presented a new error estimate for the Local Discontinuous Galerkin method applied to linear convection diffusion problem with Dirichlet boundary conditions. We have, theoretically and numerically, demonstrated an optimal order of convergence of $k + 1$, when using a suitable numerical flux. Extension to other types of boundary conditions such as Neumann boundary conditions and to higher dimensional problems will be treated in a forthcoming paper.

References

[CS98] Cockburn, B., Shu, C.W.: The Local Discontinuous Galerkin Finite Element Method for convection diffusion systems. SIAM J. Numer. Anal. **35**, (1998) 2440–2463

[CSK] Cockburn, B., Shu, C.W., Karniadakis, G.E.: An overview of the development of DG methods. This volume, Springer Verlag, Berlin, Germany,

[LR74] LeSaint, P., Raviart, P.A.: On a finite element method for solving the neutron transport equation. In: Mathematical aspects of finite elements in partial differential equations. Academic Press, C.A de Boor (Ed), Academic Press, New York (1974), 89-145